

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC602 - FOUNDATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following conditions is most critical for ensuring satisfactory performance of a shallow foundation under service loads?
A. Maximum shear strength of soil
B. Minimum factor of safety against bearing capacity failure
C. Limiting settlement within tolerable values
D. Uniform soil stratification up to great depth
Ans. C. Limiting settlement within tolerable values
- 2 For a foundation resting on normally consolidated clay, the dominant component of settlement is:
A. Immediate settlement
B. Secondary compression
C. Consolidation settlement
D. Differential settlement
Ans . C. Consolidation settlement
- 3 Which failure mode assumed in Terzaghi's bearing capacity theory is valid primarily for dense sand or stiff clay?
A. Local shear failure
B. Punching shear failure
C. General shear failure
D. Progressive failure
Ans. C. General shear failure
- 4 For an **infinite slope in cohesionless soil**, the factor of safety depends mainly on:
A. Cohesion and pore water pressure
B. Unit weight and depth of slip surface
C. Angle of internal friction and slope angle
D. Width of slope and surcharge
Ans. C. Angle of internal friction and slope angle

- 5 The **Friction Circle Method** of slope stability analysis is best described as:
 A. A purely analytical method
 B. A semi-graphical method
 C. A numerical method
 D. An empirical method
 Ans. B. A semi-graphical method
- 6 Taylor's Stability Number is most useful for analyzing:
 A. Infinite slopes in sand
 B. Anchored sheet pile walls
 C. Finite slopes in cohesive soils
 D. Rock slopes with discontinuities
 Ans. C. Finite slopes in cohesive soils
- 7 Which in-situ test provides direct estimation of allowable bearing pressure with minimum soil disturbance?
 A. Standard Penetration Test (SPT)
 B. Plate Load Test
 C. Cone Penetration Test (CPT)
 D. Pressuremeter Test
 Ans. B. Plate Load Test
- 8 SPT-N values are corrected primarily to account for:
 A. Grain size distribution
 B. Moisture content variation
 C. Soil plasticity
 D. Overburden pressure and energy efficiency
 Ans. D. Overburden pressure and energy efficiency
- 9 Core recovery ratio obtained during rock drilling mainly indicates:
 A. Strength of rock mass
 B. Degree of weathering
 C. Quality and continuity of rock core
 D. Bearing capacity of rock
 Ans. C. Quality and continuity of rock core
- 10 The **allowable bearing pressure** of soil is obtained by:
 A. Dividing ultimate bearing capacity by factor of safety
 B. Subtracting settlement from ultimate load
 C. Using load factors as per IS 456
 D. Applying partial safety factors on soil parameters
 Ans. A. Dividing ultimate bearing capacity by factor of safety
- 11 Negative skin friction in piles generally occurs when:
 A. Pile is driven into dense sand
 B. Surrounding soil settles more than the pile
 C. Uplift force acts on pile
 D. Pile tip rests on hard rock
 Ans. B. Surrounding soil settles more than the pile
- 12 Which foundation type is **most suitable** for expansive soils when swelling pressure is significant?
 A. Isolated footing
 B. Combined footing
 C. Under-reamed pile foundation
 D. Raft foundation
 Ans. C. Under-reamed pile foundation

Q2. Solve the following.

- A) Explain the necessity and objectives of soil exploration for foundation design. 6
Also explain the consequences of inadequate soil investigation.

Ans.

1. Necessity of Soil Exploration

Soil exploration is required because soil is heterogeneous and anisotropic, and foundation performance depends on subsurface conditions. It is essential to:

- To determine safe bearing capacity of soil,
- To estimate settlement (often governing design),
- To locate weak or compressible layers,
- To identify problematic soils such as expansive, collapsible, or organic soils,
- To decide suitable foundation type (shallow, raft, or pile),
- To ensure safety, serviceability, and economy of the structure.

2. Objectives of Soil Exploration

The main objectives are:

- To determine **soil stratification and thickness of layers**
- To establish **groundwater table location and variation**
- To obtain **engineering properties of soil**, such as:
 - shear strength parameters (c, ϕ ,)
 - compressibility parameters (C_c, m_v)
 - permeability (k)
- To provide **design parameters** for bearing capacity and settlement
- To identify construction difficulties like dewatering or ground improvement needs

3. Consequences of Inadequate Soil Investigation

Inadequate investigation may result in:

- unexpected **bearing capacity failure**,
- excessive or **differential settlement**,
- cracking and tilting of structures,
- wrong foundation selection,
- costly remedial measures or litigation.

- B) A 1.5 m long core barrel is used for rock drilling. The total length of sound core recovered is 0.9 m. Calculate the **Core Recovery Ratio (CRR)** and comment on **rock quality and engineering significance**. 6

Ans.

$$\text{CRR} = (\text{Length of core recovered} / \text{Core run length}) \times 100$$

$$\text{CRR} = (0.9/1.5) \times 100 = 60\%$$

- **CRR = 60%** indicates **moderate to poor rock quality**
- Suggests fractured, weathered, or jointed rock mass

Engineering Significance

- Influences choice of foundation (open vs pile or caisson)
- Affects allowable bearing pressure on rock
- Indicates need for grouting or improvement
- Important for stability of deep excavations and wells

Q3. Solve the following.

- A) Explain the three modes of bearing capacity failure of soil and state the soil conditions under which each occurs. 6

Ans.

1) General Shear Failure

- Occurs in **dense sand or stiff clay**
- Well-defined **continuous shear surfaces** extend to ground surface
- Sudden failure accompanied by **bulging of soil**
- Load–settlement curve shows a **distinct peak**

Engineering significance:

Failure is sudden and catastrophic; design is governed by bearing capacity.

2) Local Shear Failure

- Occurs in **medium dense sand or medium stiff clay**
- Shear surfaces are **partially developed**
- Settlement increases gradually
- Load–settlement curve does **not show a sharp peak**

Engineering significance:

Bearing capacity equations are modified by reducing shear strength parameters.

3) Punching Shear Failure

- Occurs in **loose sand or soft clay**
- Footing **punches into soil** without surface heave
- No clear failure surface
- Large settlement occurs even at low loads

Engineering significance:

Settlement governs design rather than shear failure.

- B) A **strip footing** of width **B = 2.0 m** is founded at depth **Df = 1.0 m** in soil having: 6
 $c=20 \text{ kPa}$, $\phi=30^\circ$, $\gamma=18 \text{ kN/m}^3$,
 Use **Terzaghi's equation** with factors for $\phi=30^\circ$
 $N_c=30$, $N_q=18$, $N_\gamma=22$
 Find **ultimate bearing capacity** q_u and **safe bearing capacity** for **FOS = 3**.

Ans.

Terzaghi (strip footing):

$$q_u = cN_c + \gamma D_f N_q + 0.5\gamma B N_\gamma$$

1. $cN_c = 20 \times 30 = 600 \text{ kPa}$
2. $\gamma D_f N_q = 18 \times 1.0 \times 18 = 324 \text{ kPa}$
3. $0.5\gamma B N_\gamma = 0.5 \times 18 \times 2.0 \times 22$
 First $0.5 \times 18 = 9$
 $9 \times 2.0 = 18$
 $18 \times 22 = 396 \text{ kPa}$

$$q_u = 600 + 324 + 396 = 1320 \text{ kPa}$$

Safe:

$$q_{safe} = \frac{q_u}{3} = \frac{1320}{3} = 440 \text{ kPa}$$

Q4. Solve Any Two of the following.

- A) Explain **collapsible soils** and **corrosive soils**. Discuss their **engineering problems** and **foundation design considerations**. 6

Ans.

1. Collapsible Soils

Collapsible soils are **loose, unsaturated soils** that undergo **sudden settlement upon wetting** under load.

Common examples:

- Loess
- Aeolian silts

Problems:

- Sudden excessive settlement
- Cracking and structural distress

Remedial measures:

- Pre-wetting and compaction
- Soil replacement
- Deep foundations bypassing collapsible layer

2. Corrosive Soils

Corrosive soils contain **chemicals (sulphates, chlorides, acids)** that attack concrete and steel.

Problems:

- Corrosion of reinforcement
- Loss of pile cross-section

Preventive measures:

- Use of sulphate-resistant cement
- Protective coatings on piles
- Increased cover to reinforcement
- Reduced durability

- B) State the assumptions and limitations of rigid design analysis of shallow foundations. Mention situations where rigid analysis is applicable. 6

Ans.

Assumptions of Rigid Analysis

- Footing behaves as a **rigid body**
- Soil pressure distribution is **linear**
- No bending deformation of footing
- Contact between soil and footing is perfect
- Settlement is uniform or linearly varying

Limitations

- Not valid for large rafts or flexible footings
- Does not consider soil–structure interaction
- May underestimate bending moments in flexible slabs

Applications

- Isolated footings
- Combined and strap footings
- Small rafts on stiff soils

- C) A building foundation exerts a **net pressure of 65 kPa** on expansive soil. The soil has a **measured swelling pressure of 40 kPa**. 6

1. Check whether the foundation is **safe against heave**.
2. State whether **special foundation measures** are required.

Ans.

Step 1: Compare pressures

- Applied pressure = 65 kPa
- Swelling pressure = 40 kPa

Since:

65 kPa > 40 kPa

The applied pressure is **greater than swelling pressure**

Step 2: Safety Assessment

- Soil swelling will be **suppressed**
- No upward heave expected

Step 3: Engineering Judgment

- Foundation is **safe against heave**
- However, **moisture control measures** are still recommended to prevent differential movement

Q5. Solve Any Two of the following.

A) Classify **pile foundations** based on

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- (a) function,
- (b) material, and
- (c) method of installation.

Ans.

(a) Classification Based on Function

- **End-bearing piles** – transfer load to hard stratum/rock
- **Friction piles** – load resisted by skin friction
- **Tension piles** – resist uplift forces
- **Compaction piles** – densify loose soil
- **Sheet piles** – retain soil/water (not load-bearing)

(b) Classification Based on Material

- Timber piles <https://www.batuonline.com>
- Concrete piles (precast & cast-in-situ)
- Steel piles
- Composite piles

(c) Classification Based on Installation

- Driven piles
- Bored piles
- Driven-and-cast-in-situ piles

B) A pile of diameter **0.4 m** and length **12 m** is embedded in clay.

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Given:

- Undrained shear strength $c_u = 60$ kPa
- Adhesion factor $\alpha = 0.6$
- Bearing factor $N_c = 9$

Calculate **ultimate load capacity of pile**. (Neglect overburden at pile tip.)

Ans.

Step 1: End Bearing Capacity

$$Q_b = A_p \times N_c \times c_u = (\pi/4) \times 0.4^2 \times 9 \times 60 = 68.04 \text{ kN}$$

Step 2: Skin Friction

$$Q_s = A_s \times T = \pi d L \times \alpha \times c_u = \pi \times 0.4 \times 12 \times 0.6 \times 60 = 542.9 \text{ kN}$$

Step 3: Ultimate Capacity

$$Q_u = Q_b + Q_s = 68.04 + 542.9 = 610.94 \text{ kN}$$

- C) Classify **caisson foundations** and explain **open caisson construction with difficulties and remedial measures**. 6

Ans.

Types of Caissons

- Open caisson
- Box caisson
- Pneumatic caisson

Open Caisson Construction

- Sinking by excavation inside cutting edge
- Self-weight causes penetration

Difficulties

- Tilting
- Shifting
- Sand blowing
- Excessive skin friction

Remedial Measures

- Kentledge loading
- Water jetting
- Eccentric excavation correction
- Air or water balancing

Q6. Solve Any Two of the following.

- A) Explain the types of slope failures commonly encountered in soil slopes. 6

Ans

1. Rotational (Circular) Failure

- Occurs in **homogeneous cohesive soils**
- Slip surface is **circular**
- Common in earth dams and embankments

2. Translational Failure

- Slip occurs along **planar surface**
- Common in stratified soils and slopes with weak layers

3. Infinite Slope Failure

- Slope length is very large compared to depth
- Failure surface is **parallel to ground surface**

4. Compound Failure

- Combination of circular and planar surfaces

Identification of failure type helps in selecting the **appropriate analysis method**.

B) An infinite clay slope has:

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- cohesion $c=25$ kPa
- unit weight $\gamma=18$ kN/m³
- depth of slip plane $z=2$ m
- slope angle $\beta=30^\circ$

Find the **factor of safety**.

Ans.

$$F = \frac{c}{\gamma z \sin \beta \cos \beta}$$

$$\sin 30^\circ = 0.5, \quad \cos 30^\circ = 0.866$$

$$F = \frac{25}{18 \times 2 \times 0.5 \times 0.866}$$

$$F = \frac{25}{15.588} = 1.60$$

C) A slope in clay has:

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- height $H=10$ m
- cohesion $c=40$ kPa
- unit weight $\gamma=20$ kN/m³

From Taylor's chart, **stability number $N_s=0.18$**

Find the **factor of safety**.

Ans.

$$N_s = \frac{c}{\gamma H F}$$

$$F = \frac{c}{N_s \gamma H}$$

$$F = \frac{40}{0.18 \times 20 \times 10} = \frac{40}{36} = 1.11$$

*** End ***